



Final Report for saudi law نظام العمل

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Introduction:

The Saudi Labor Law is considered one of the most important regulations that ensure a fair and motivating work environment for all employees in the Kingdom, as it aims to protect workers' rights and regulate the relationship between employees and employers, thereby promoting workplace justice. Despite its importance, this law consists of a set of legal rules written in specialized language and filled with technical terms. Here lies the problem: the difficulty of quickly accessing the required information, as non-specialists in law—such as employees, employers, and others—face major challenges when searching for a specific legal article in a short time.

Therefore, this project aims to transform the articles of the Saudi Labor Law into a structured Question–Answer (Q&A) dataset that enables various beneficiary groups to quickly access legal texts. This highlights the importance of presenting these legal materials in a way that is clear, organized, and easily searchable for all users.

Research Question: How can the articles of the Saudi Labor Law be transformed into a structured Question–Answer dataset that supports quick access to legal information and serves various beneficiary groups?

Data sources:

- 1- Bureau of Experts at the Council of Ministers
- 2- Istitlaa Platform
- 3- Qiwa Platform
- 4- Master's Theses from King Saud University Digital Library

Source1:

Description: The official text of the Saudi Labor Law, collected from the official website of the Bureau of Experts at the Council of Ministers. This text represents the legal framework that regulates the relationship between employees and employers in the Kingdom of Saudi Arabia.

Data Type: Unstructured text data, as the file consists of lengthy legal articles in Arabic provided in PDF format.

Potential Biases:

1-Representation: The law formally covers workers and employers, but it may not fully reflect the situations of informal or undocumented workers.

2-Measurement: The text is accurate since it is taken directly from the official source, but the specialized legal language may make access to information slower or more difficult for non-specialists.

3-Historical Bias: The text reflects only the most recent amendments and does not include previous versions of the law or rights that may have changed over time.

Sorce-link: <https://www.boe.gov.sa/>

Source2:

Description: Draft amendments to the Implementing Regulations of the Saudi Labor Law, published on the Istitlaa (Public Consultation Platform). The document was prepared and issued by the Ministry of Human Resources and Social Development (MHRSD) as part of Vision 2030 initiatives to enhance labor market efficiency, regulate employment practices, and gather public and stakeholder feedback before final approval.

Data Type: Semi-structured data. The core file is provided in Arabic in PDF format with regulatory articles and proposed amendments. In addition, the platform itself provides structured metadata such as the consultation period, target groups, responsible authority (MHRSD), and the project status (active/closed).

Potential Biases:

1- Representation: The draft reflects the Ministry's official proposal, but it may not represent the final legally binding regulation. It also primarily addresses the formal labor market and may not cover informal or undocumented workers.

2- Measurement: The content is accurate as it comes directly from the Ministry, but the technical and legal language may pose accessibility challenges for non-specialists.

3- Historical Bias: The draft captures only the most recent proposed changes and does not include previous versions of the regulation or a full record of earlier labor law practices.

Source-link: <https://istitlaa.ncc.gov.sa/>

Source3:

Description: Qiwa is the official digital platform under the Ministry of Human Resources and Social Development (MHRSD) that provides integrated electronic services for employers and workers in the private sector. The platform includes access to labor law content such as the main chapters of the Saudi Labor Law, practical guidelines, and Q&A on its implementation, in addition to automated services like employment contracts, company regulations, and work permits.

Data Type: Semi-structured text data. While the labor law is provided as official structured chapters and articles, the platform also offers practical interpretations in the form of Q&A and interactive guidance. In addition, metadata such as service categories, target users (workers/employers), and update timestamps are included.

Potential Biases:

1- Representation: The platform mainly focuses on the private sector and on users registered in the system; it may not fully reflect the realities of informal workers or sectors not covered by Qiwa services.

2- Measurement: The data is authoritative since it is maintained by MHRSD, but some of the content is presented as practical Q&A, which may simplify or omit complex legal nuances.

3- Historical Bias: The platform reflects the current law and updated practices. Older versions of the law or outdated interpretations are not preserved or displayed, which may limit longitudinal analysis.

Source-link: <https://qiwa.sa/>

Source4:

Description: Master's theses containing analytical studies on Saudi labor law, such as termination rules, training contracts, and remote work contracts.

Data Type: Text data, unstructured because the theses presents content as pages of paragraphs.

Title Of These Two Master's Theses:

1- "Provisions of Remote Work Contracts under the Saudi Labor Law" By: Jana Khaled Mohamed Al-Harbi & Dr. Amal Mohamed Shalaby Khodraj (2025).

2- "Termination Provisions in Training and Qualification Contracts under the Saudi Labor Law" By: Taqiyah Ali Rashwan & Dr. Al-Hanouf Al-Salmi (2024).

Potential Biases:

1- **Representation:** Covers academic legal cases; may not represent daily practical cases for all groups.

2- **Measurement:** Accurate information but analytical and unofficial compared to government sources.

3- **Historical Bias:** Relatively recent, but limited to the academic study and selected research areas.

Source: PDF Files.

King Saud University
College of Computer and Information Sciences
Department of Information Technology
IT 362: Principles of Data Science
Project – final report



كلية علوم الحاسب والمعلومات
قسم تقنية المعلومات

Source	Observations	Features	Data Types
Bureau of Experts at the Council of Ministers	245	Section Chapter Article Text Status Source	String String String Text String String
Istitlaa Platform	16	Article Current_Text Proposed_Text URL	String text text String
Qiwa	245	URL Content	String Text
Master's Theses from King Saud University Digital Library	2	Thesis_Title Author_Name Year	String String Integer

Objectives:

The main objective of this project is to develop a structured Question–Answer (Q&A) dataset from the Saudi Labor Law that can generate meaningful insights and support practical legal tasks. This dataset will provide students, researchers, and practitioners with a simplified Q&A format that improves comprehension and facilitates effective legal review

In addition, the dataset will enable thematic analysis of the law’s content, such as identifying the focus of articles on contracts, wages, leaves, termination, and disciplinary actions. It will also serve as a foundation for AI applications, supporting the training of language models to create legal assistants and intelligent search systems.

Moreover, the dataset can assist HR managers and legal professionals in addressing real-world questions—for example, probation periods, severance pay, or working hours—by linking each answer directly to official articles. Finally, it will promote transparency and compliance by ensuring that every answer remains grounded in the official legislative text, guaranteeing both accuracy and reliability.

Method:

We will begin by collecting the official Saudi Labor Law text from Qiwa.sa, the Ministry of Human Resources and master's theses from King Saud University Digital Library websites using Selenium, since these sources provide the most reliable and up to date version of the Saudi Labor law. The raw text gathered from these sources will be saved in a CSV file containing the extracted content.

Once the data is collected, we will segment the labor law into individual articles or sections to maintain clarity and preserve the original legal structure. At this stage, we will also design a clear Q&A format.

To transform the raw legal text into a Q&A dataset, we will use large language models (LLMs) to generate candidate questions and answers for each article. The focus will be on practical themes such as contracts, wages, probation periods, termination, leaves, and disciplinary actions, since these are the topics most relevant for both legal professionals and students. Importantly, we will ensure that the answers produced by the LLM remain grounded in the official legislative text so that accuracy and reliability are not compromised.

After generating the Q&A pairs, we will validate the results through manual review. This step will allow us to confirm the legal accuracy of the dataset and correct or remove any outputs where the model has misinterpreted the law.

Finally, we will consolidate the validated Q&A pairs into a structured dataset that can serve multiple purposes: as a study aid for students, as a resource for thematic analysis of Saudi labor law, and as a foundation for building AI applications such as legal assistants or intelligent search systems.

Phase2 :

In This phase presents the results of the Exploratory Data Analysis (EDA) conducted for both primary and secondary datasets related to the Saudi Labor Law. The purpose of this analysis is to better understand the content and structure of various legal and public data sources that reflect different perspectives on labor regulations in Saudi Arabia. The analysis aims to identify key patterns, relationships, and differences across four main data sources: the official labor law from the Board of Experts (BOE), public consultation data from the Istitlaa platform, explanatory articles from the Qiwa platform (Qiwa Articles), and user-generated questions from the Qiwa FAQ section. These datasets together provide a comprehensive view of how labor law information is presented, interpreted, and discussed across official and public platforms. This EDA was performed after completing the data processing and cleaning phase to ensure the accuracy and reliability of the findings. A combination of descriptive statistics, textual visualizations, and comparative charts was used to uncover meaningful trends and thematic similarities between the datasets. The results of each dataset were analyzed and presented separately to highlight their unique characteristics, potential biases, and target audiences. Finally, the report includes a comparative overview and a summary of the main insights and new hypotheses derived from the analysis.

Primary Data:

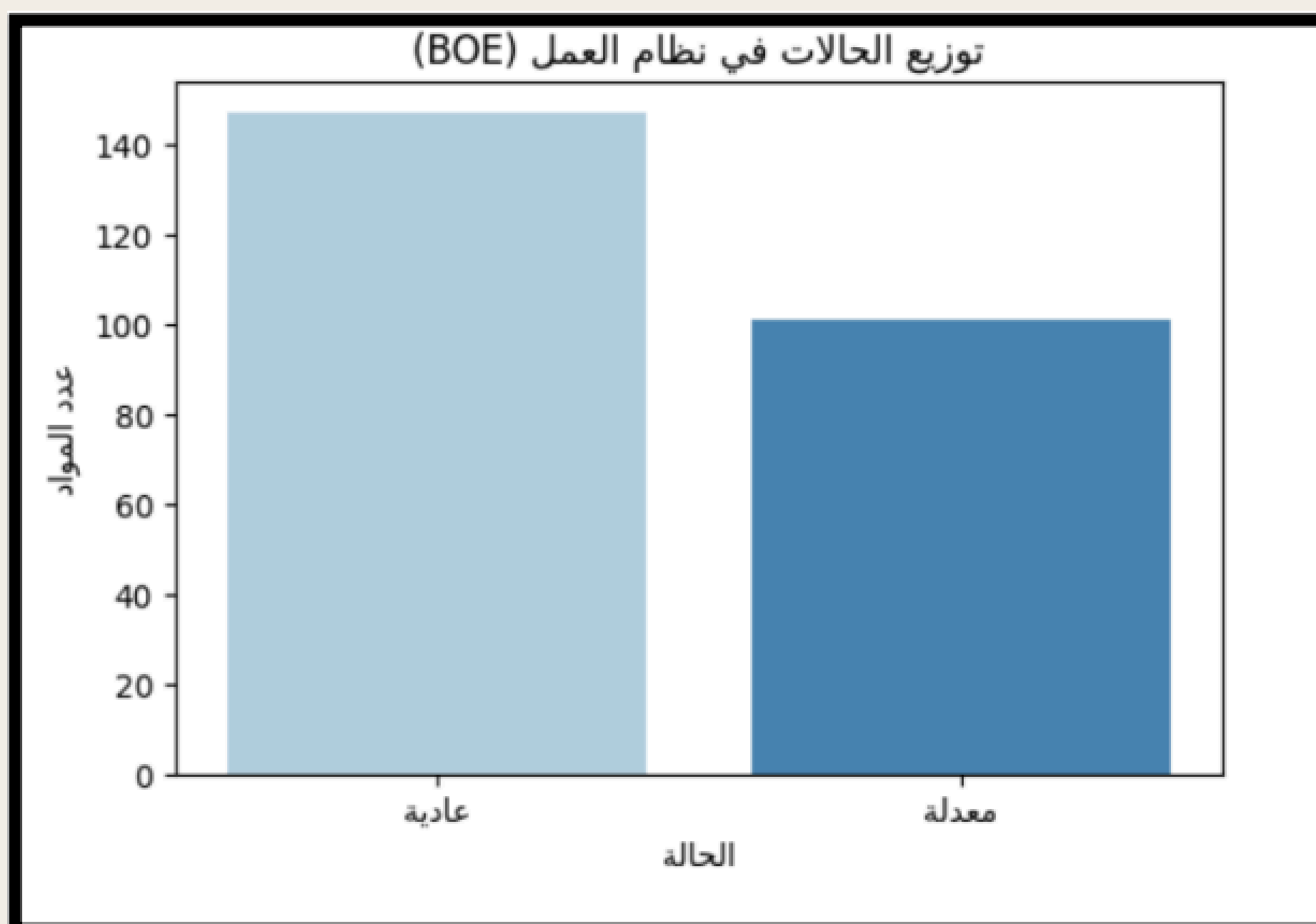
BOE Brief description:

The dataset was obtained from the Saudi Council of Experts at the Council of Ministers, a specialized advisory body responsible for preparing, reviewing, and amending laws and regulations in Saudi Arabia. It includes the latest version of the Saudi Labor Law with all its articles and amendments

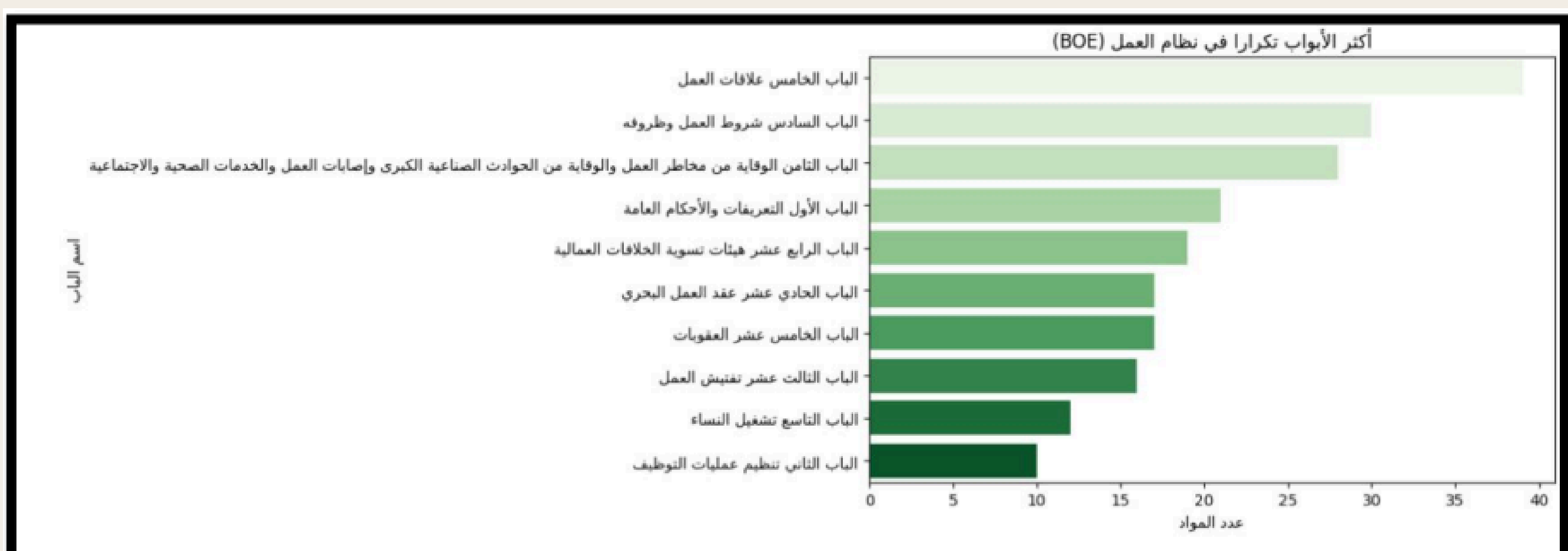
Dataset overview:

```
◆ BOE Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 248 entries, 0 to 247
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Section     248 non-null   object
1   Chapter     248 non-null   object
2   Article     248 non-null   object
3   Text        248 non-null   object
4   Status      248 non-null   object
5   Source      248 non-null   object
dtypes: object(6)
memory usage: 11.8+ KB
```

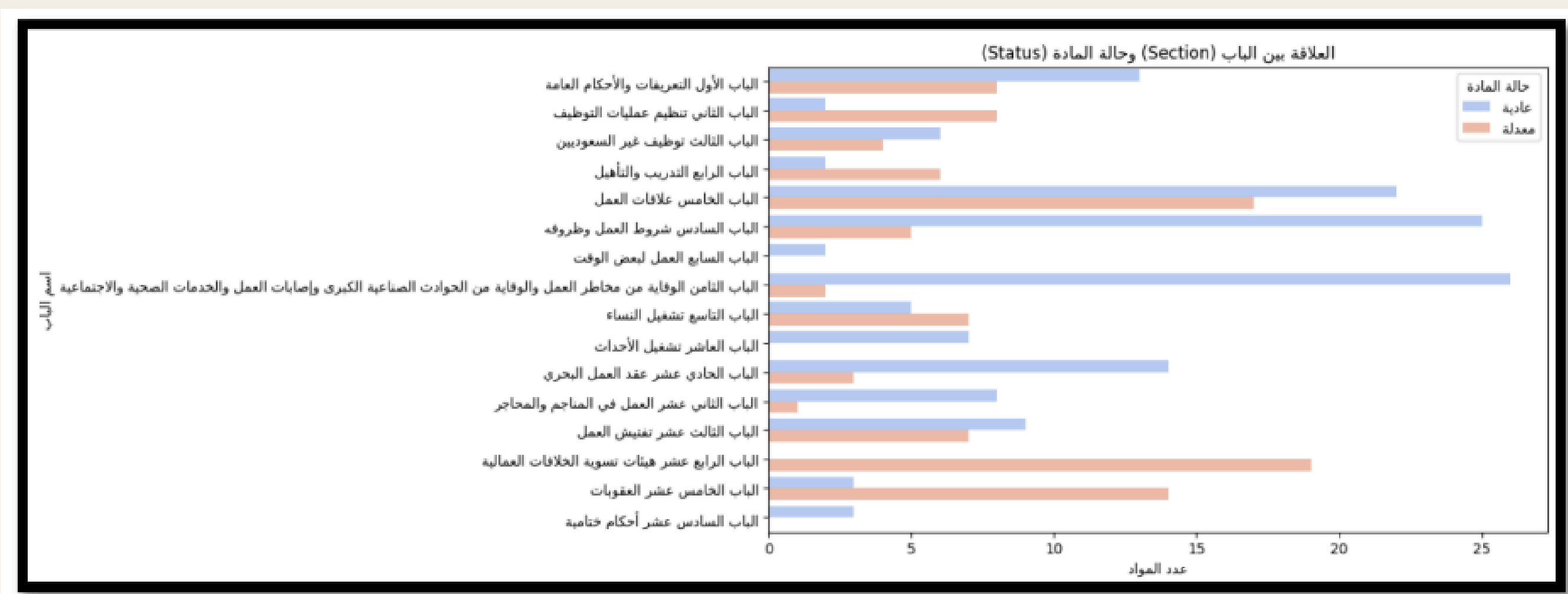

Status distribution in BOE:



The most frequent sections in the BOE Labor Law:



Relationship between section and article status:



BOE Insights:

The BOE dataset reveals that legislative modifications are concentrated within specific sections of the Saudi Labor Law. Certain sections show a higher frequency of updated or amended articles, indicating that regulatory efforts primarily focus on a limited set of labor-related themes rather than being uniformly distributed across all topics.

Secondary Data:

A-First Source: Istitleaa

Brief description: The dataset was collected from the Saudi “Isttelaa” platform, which gathers public and institutional feedback on draft laws and regulations. It focuses on the latest Saudi Labor Law.

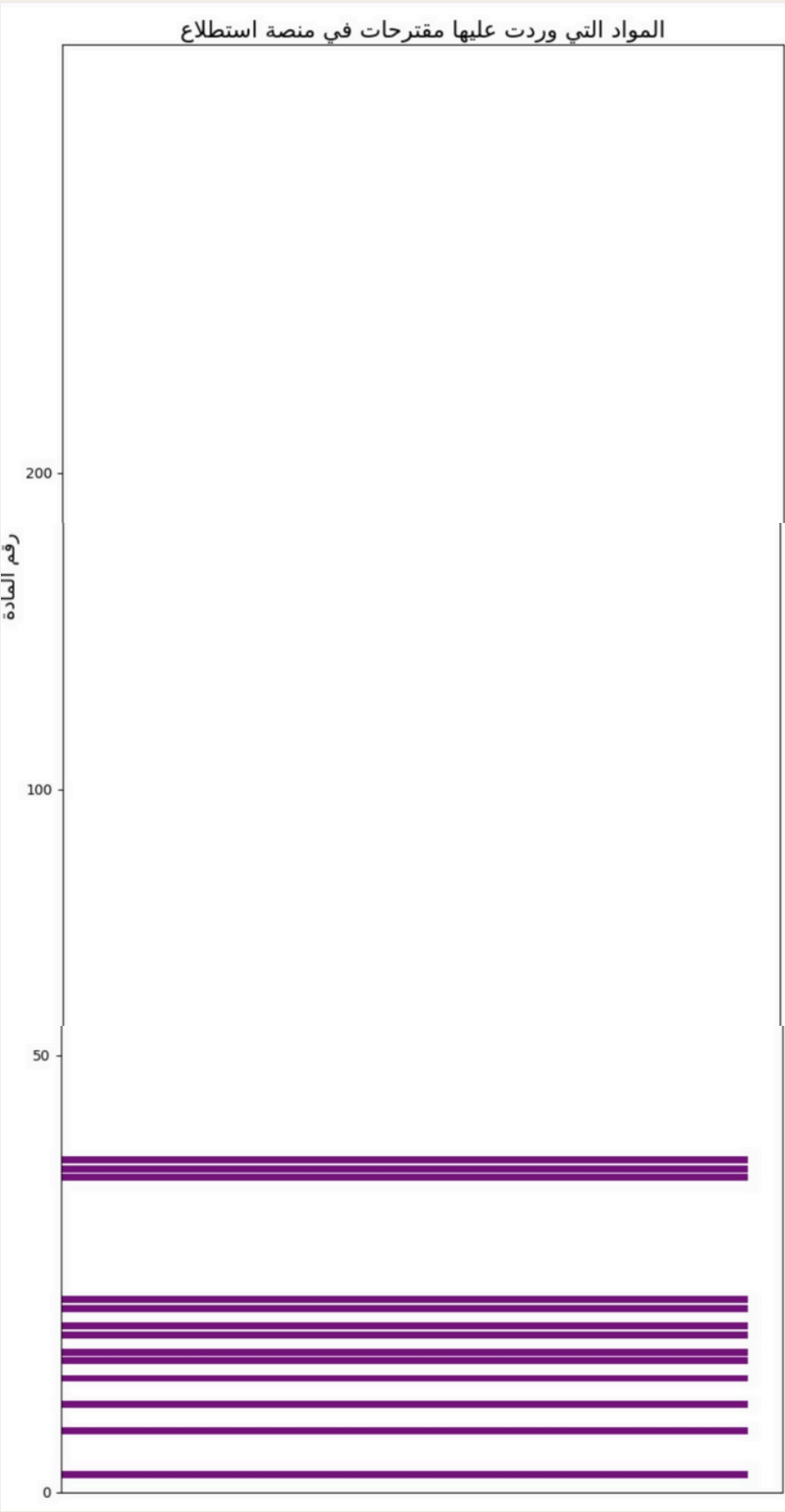
Bias:

Although the data is sourced from an official government consultation platform, the analyzed texts are public opinions, comments, and suggestions rather than binding legal texts. This introduces an Opinion Bias, meaning the content includes a high degree of personal judgment and estimations which the project aims to minimize in the new regulations and penalties table.

Dataset overview:

```
◆ Istitlaa Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Article         16 non-null    object
1   Current Text    16 non-null    object
2   Proposed Text   16 non-null    object
3   URL             16 non-null    object
dtypes: object(4)
memory usage: 644.0+ bytes
```

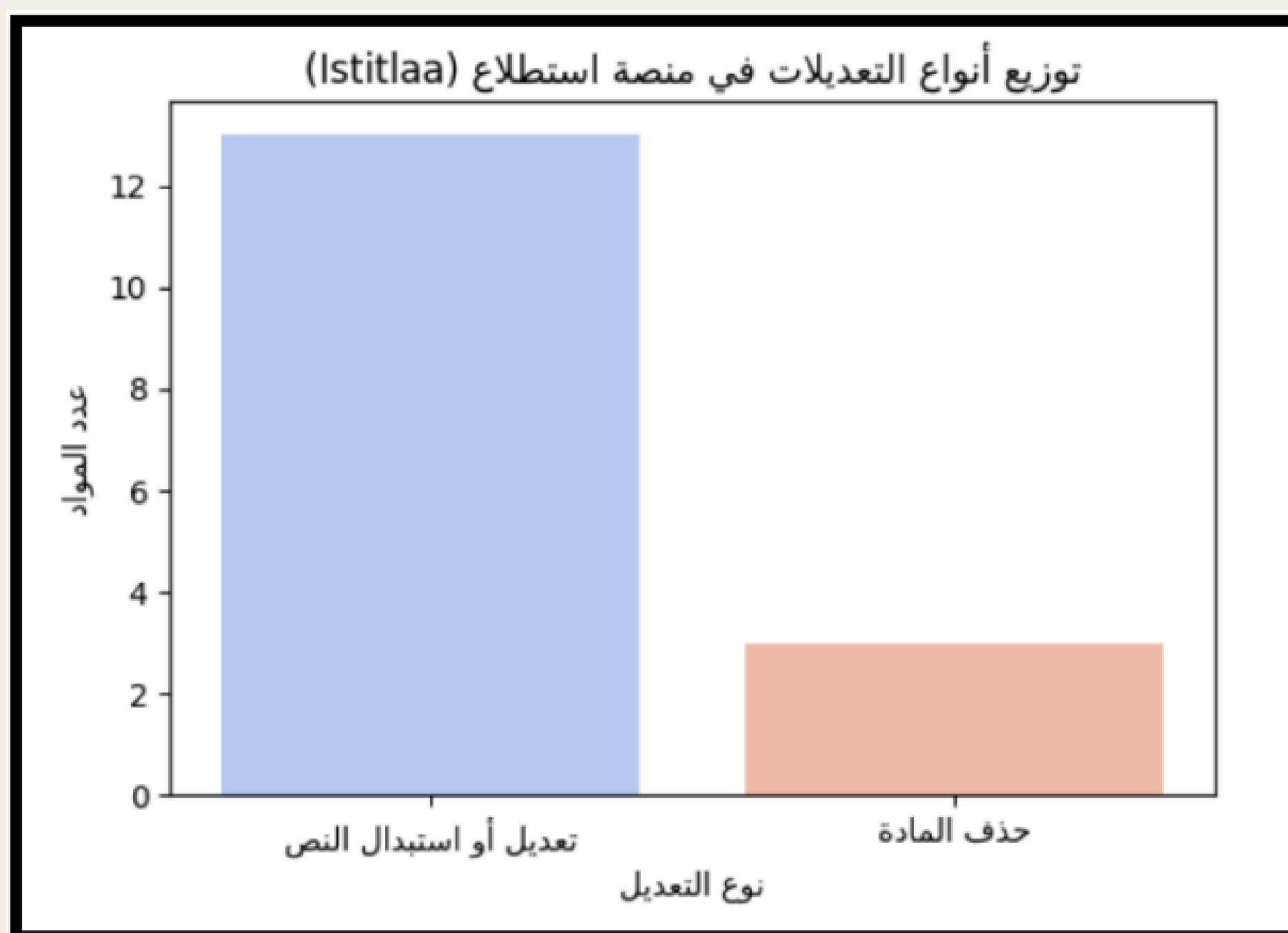
Article proposal frequency:



Categorize proposed changes:

	Article	Change_Type
0	المادة 2	حذف المادة
1	المادة 7	تعديل أو استبدال النص
2	المادة 10	تعديل أو استبدال النص
3	المادة 13	تعديل أو استبدال النص
4	المادة 15	تعديل أو استبدال النص

Distribution of change types:



Istitlaa Insights:

The Istitlaa dataset demonstrates that most public feedback emphasizes practical and actionable changes—such as requests for updates, training policies, or procedural modifications. The frequency distribution of proposed changes highlights that public participants focus on improving clarity and applicability in specific areas rather than suggesting broad or theoretical amendments.

B-Second Source: Qiwa articles

Brief description:

The dataset focuses on the Saudi Labor Law from the “Qiwa” platform, covering its articles and related explanatory materials for employers. While the platform offers additional tools and examples, this dataset specifically includes the core labor law content.

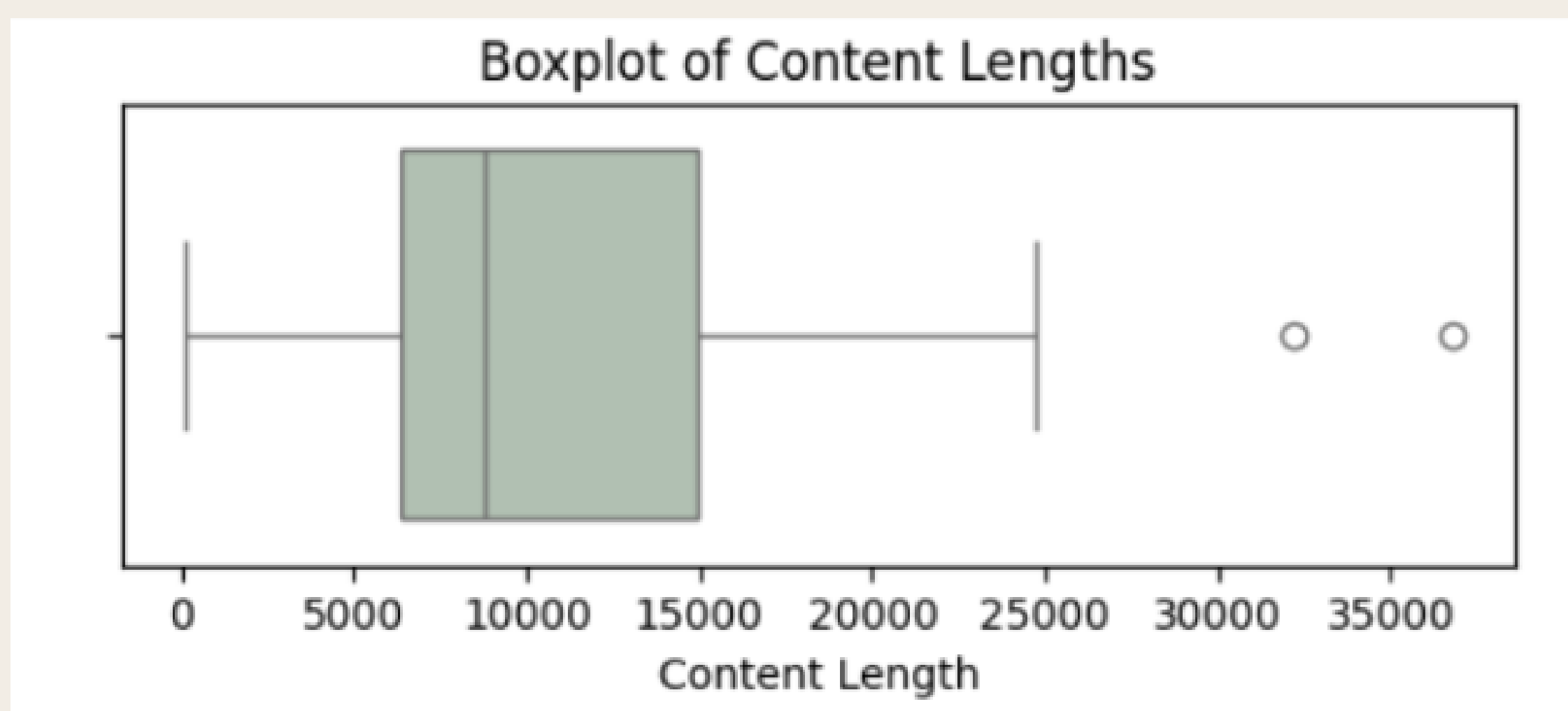
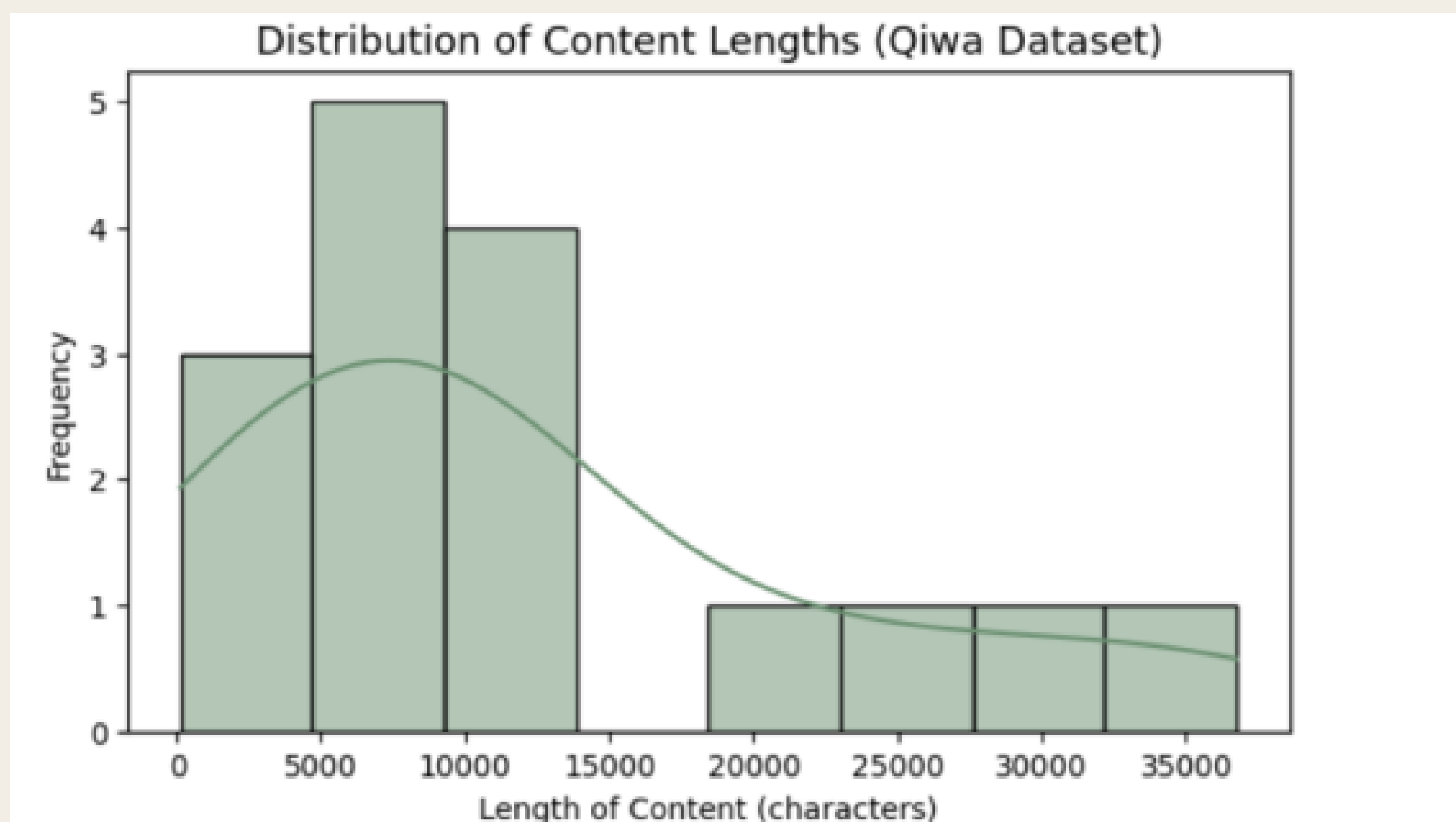
Bias:

These articles are designed as comprehensive and educational texts targeting employers, which introduces an Interpretation Bias toward presenting an idealized, systematic view of the labor law. They may not sufficiently cover the specific details of violations and penalties which is the core focus of this project's updated table. The content, therefore, requires supplementation with data that offers more concrete specifications.

Dataset overview:

```
=== Qiwa (Cleaned) ===  
Shape: (16, 2)  
  
Columns:  
['URL', 'Content']  
  
Data types:  
URL      object  
Content   object  
dtype: object  
  
Missing values per column:  
URL      0  
Content   0  
dtype: int64  
  
Approx. duplicate rows: 0
```

Content length distribution:



Qiwa Articles Insights:

The Qiwa Articles dataset contains lengthy and detailed texts that reflect a formal, explanatory tone. The frequent appearance of legal and procedural terms indicates that this content serves an educational purpose, offering systematic interpretations of labor law articles rather than discussing penalties or real-world applications.

C-Third Source: Qiwa FAQ

Brief description:

The FAQ part of Qiwa platform.

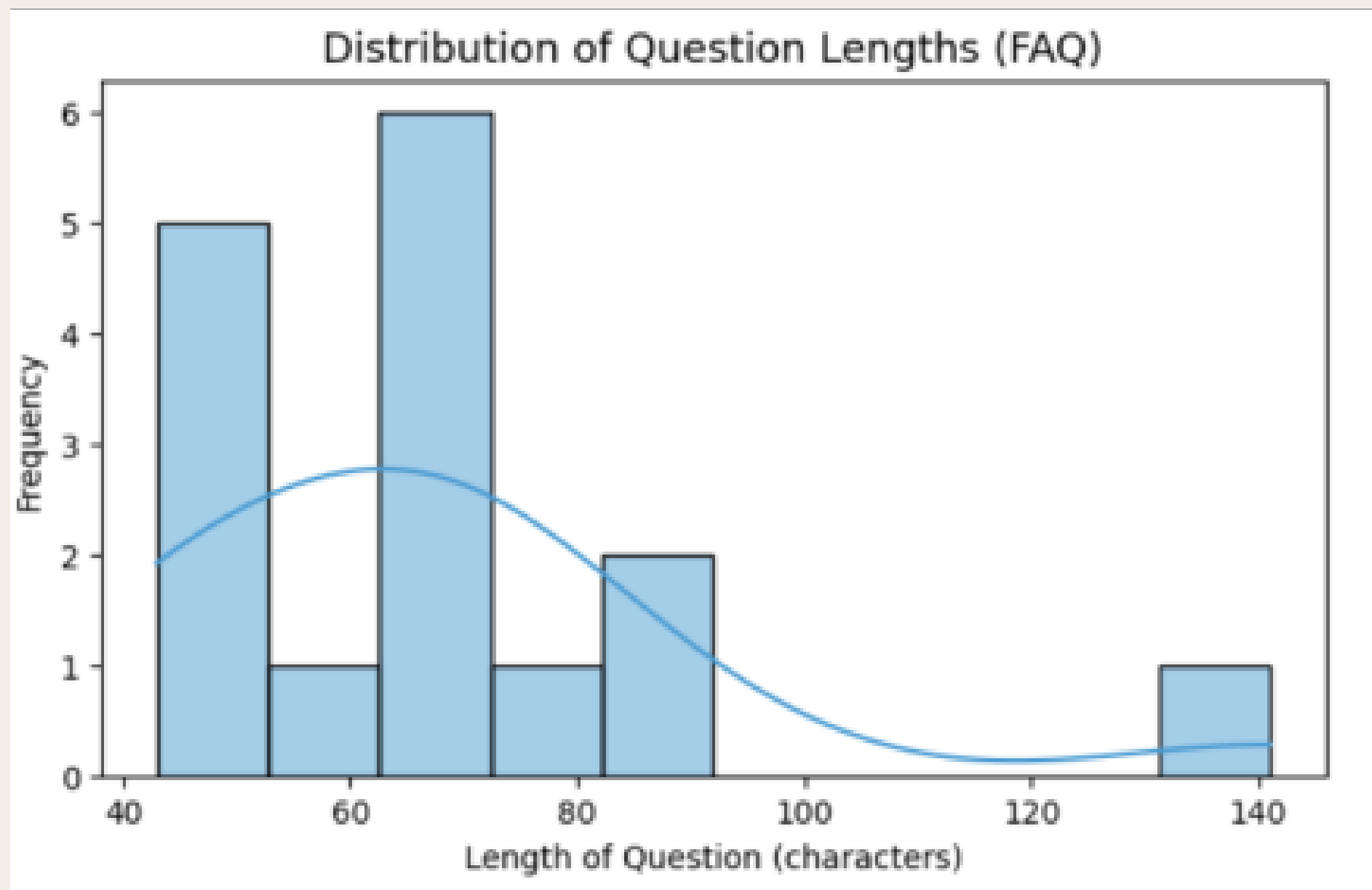
Bias:

This data reflects the most common questions raised by Qiwa platform users. Consequently, it carries a Usage Bias, focusing heavily on routine platform usage and practical daily issues, potentially neglecting less common or highly complex legal topics relevant to the broader labor law.

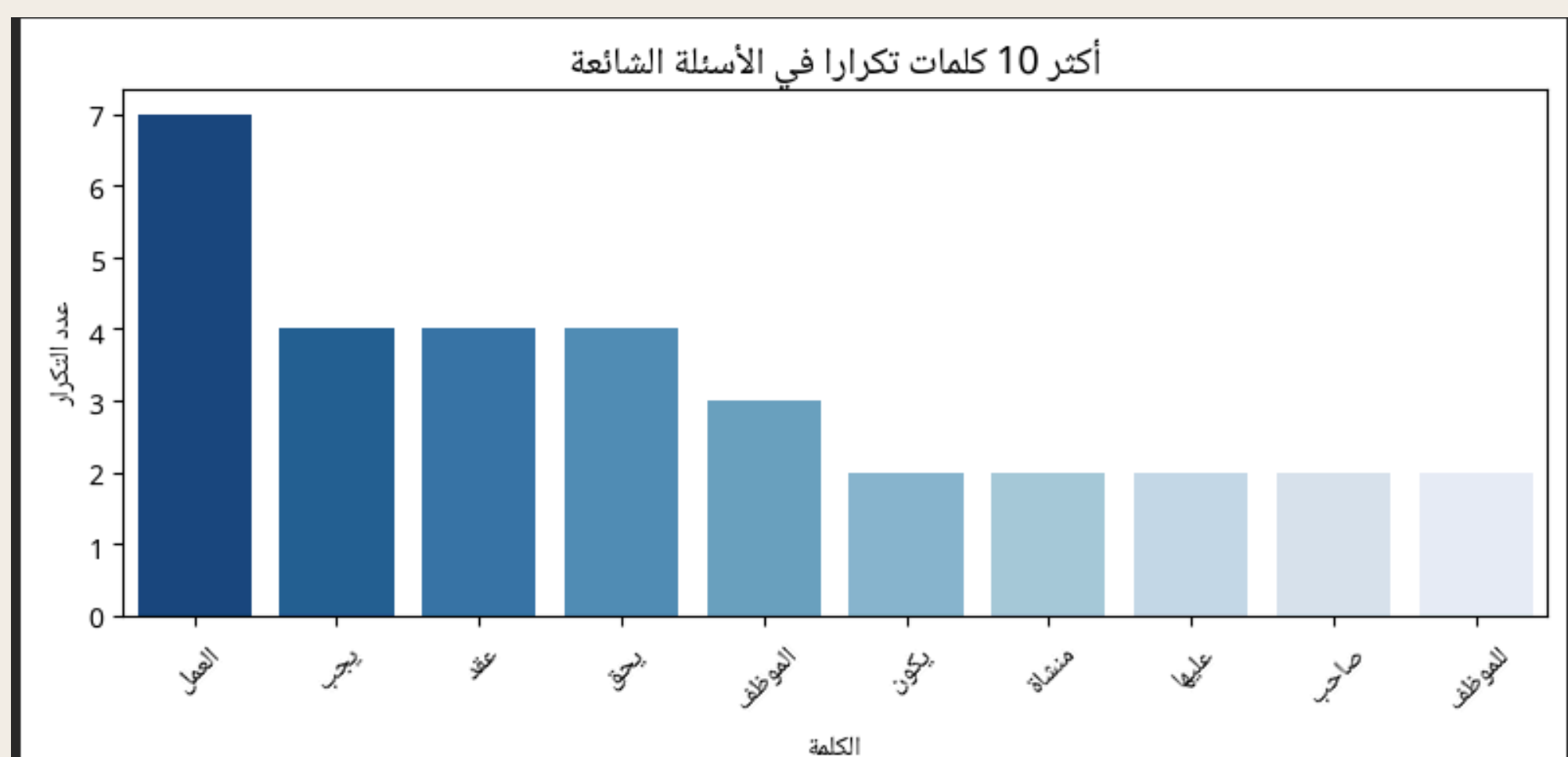
Dataset overview:

```
=== FAQ (Labor Law FAQ) ===  
Shape: (16, 2)  
  
Columns:  
['Question', 'Answer']  
  
Data types:  
Question    object  
Answer      object  
dtype: object  
  
Missing values per column:  
Question    0  
Answer      0  
dtype: int64  
  
Approx. duplicate rows: 0
```


Distribution of question lengths:



Top frequent words:



Word cloud:

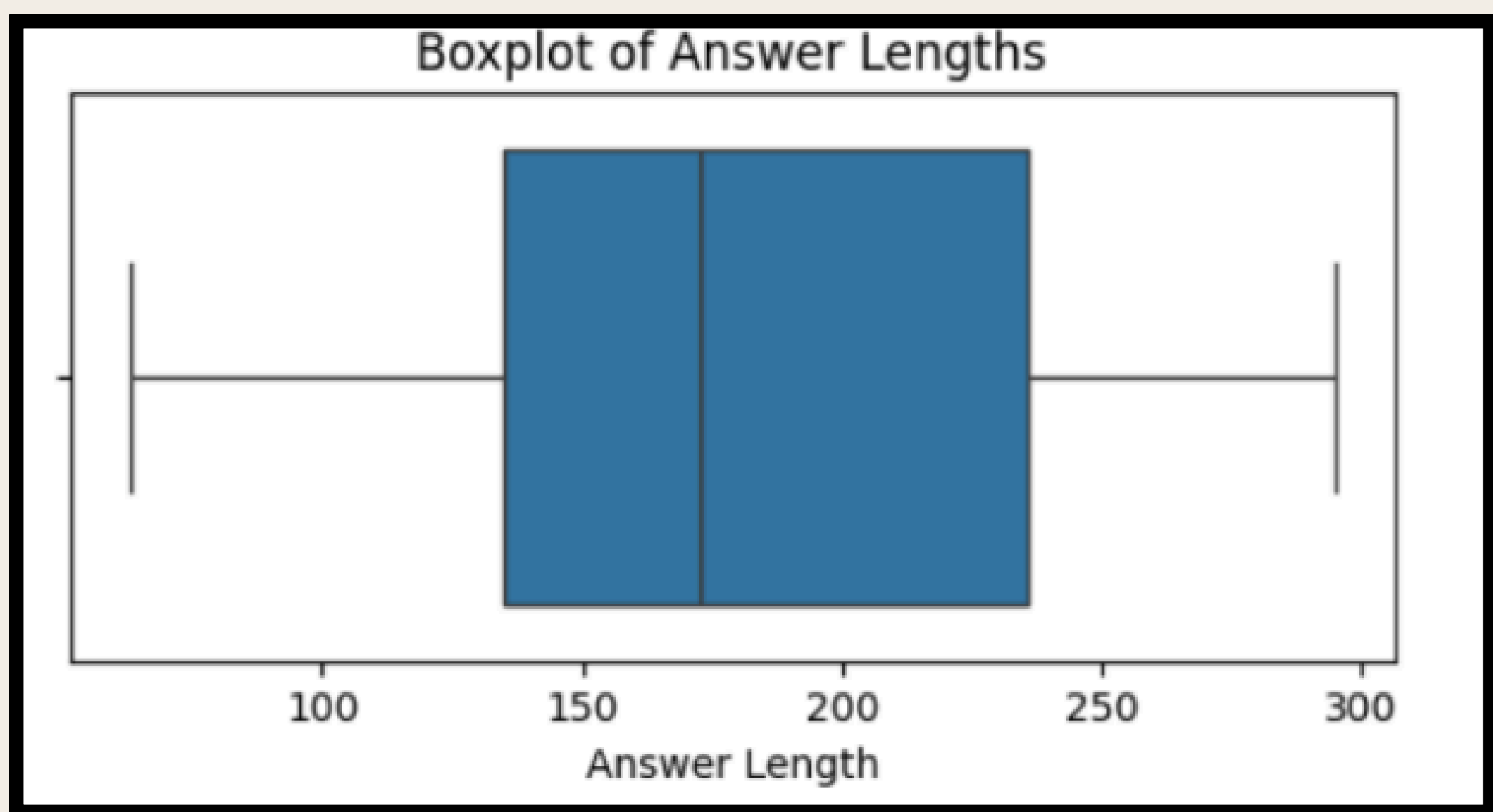
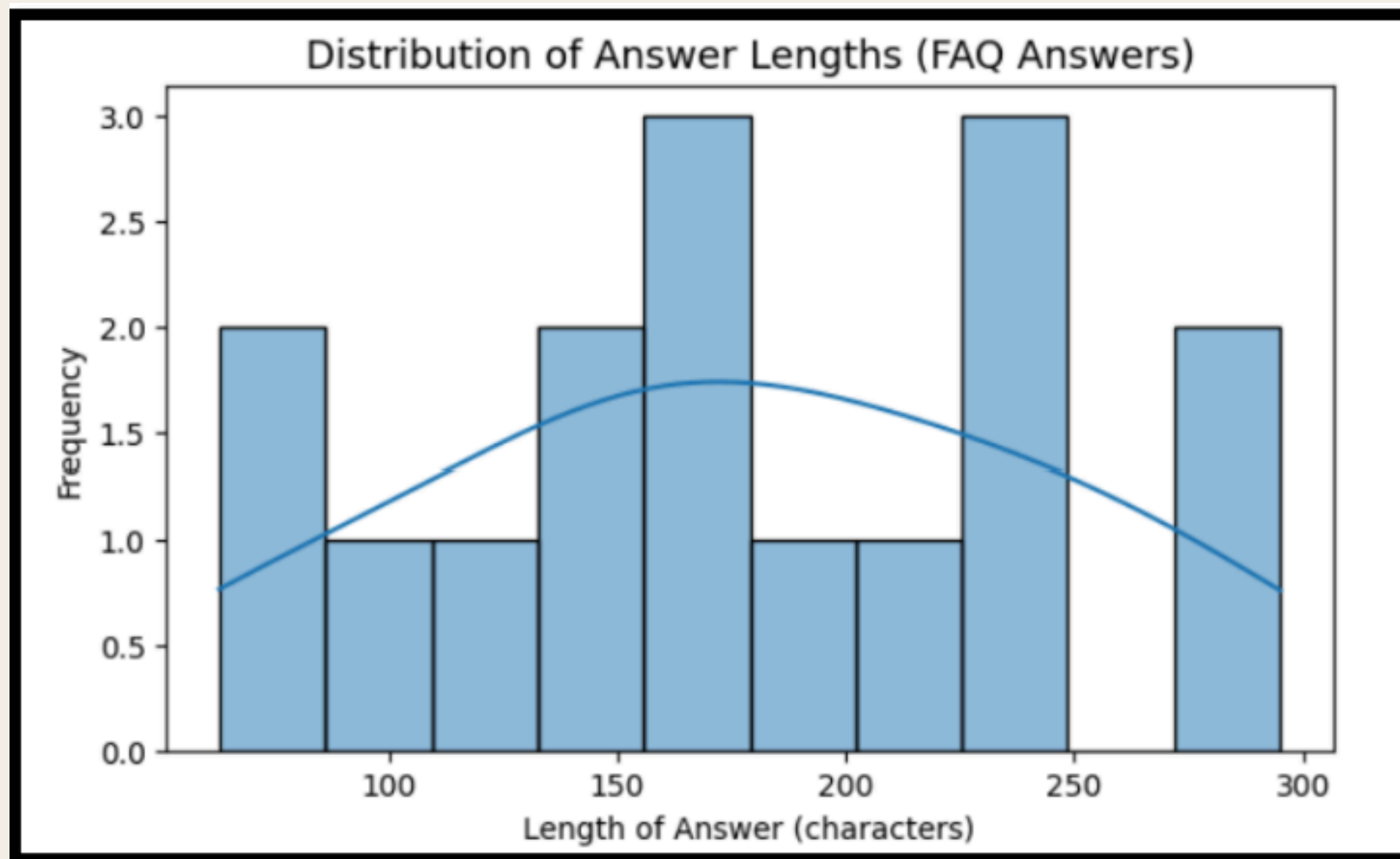


N-gram analysis bigrams & trigram:

Top 10 Bigrams:		
		Bigram Count
136		4 هل يحق
73	3	عقد العمل
154	3	يجب أن
107	3	ما هي
2	2	أن تتوفر
23	2	التي يجب
100	2	لصاحب العمل
70	2	صاحب العمل
54	2	تتوفر في
92	2	في عقد

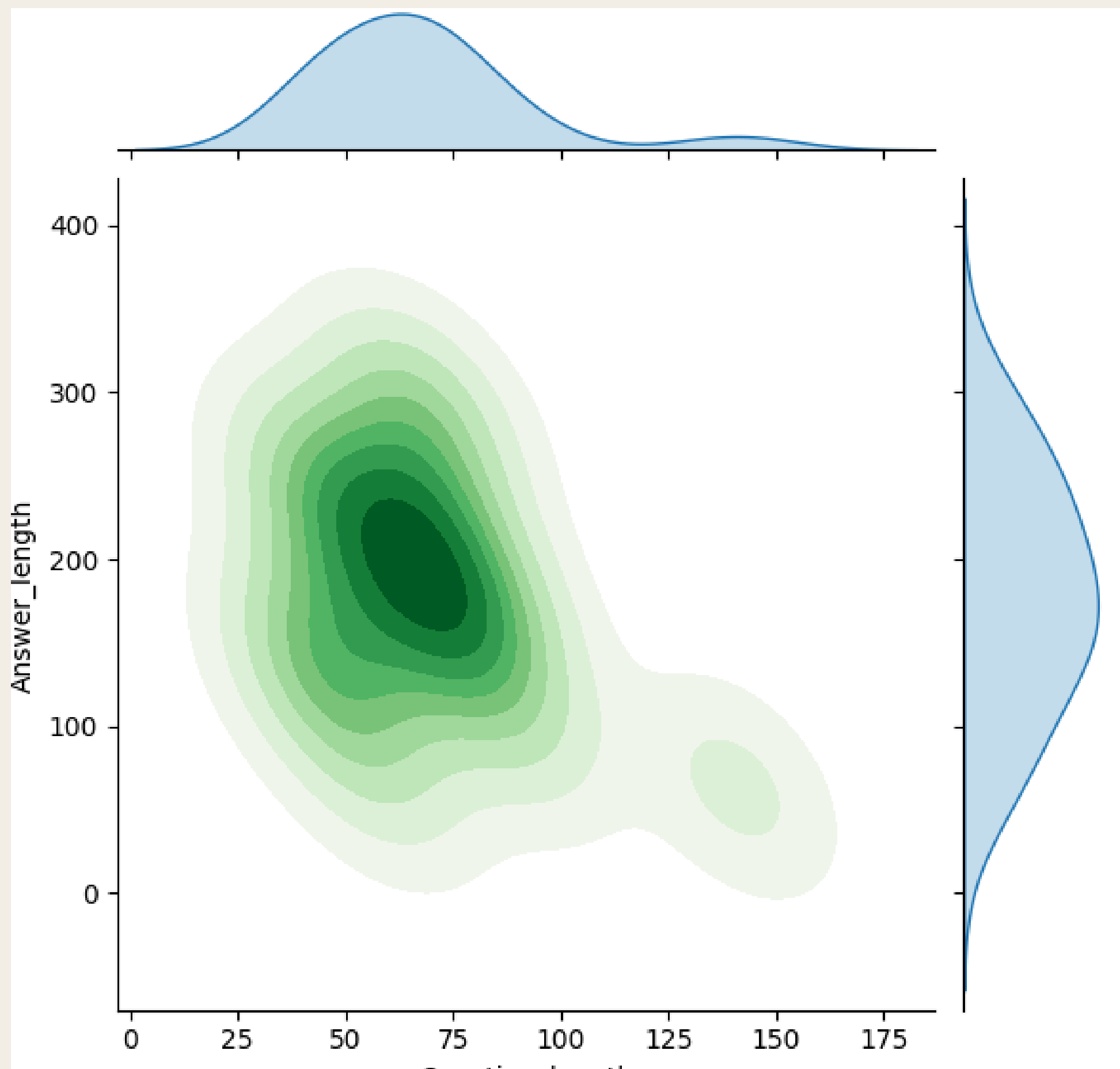
Top 10 Trigrams:		
		Trigram Count
1	2	أن تتوفر في
22	2	التي يجب أن
149	2	يجب أن تتوفر
86	2	في عقد العمل
130	2	هل يحق لصاحب
152	2	يحق لصاحب العمل
50	2	تتوفر في عقد
3	1	أن ينقل موظف
0	1	أخرى دون موافقة
9	1	إلى غير محدد

Histogram of answer lengths:



Qiwa FAQ Insights: The Qiwa FAQ dataset consists of short, focused entries designed to address common user inquiries. The frequent terms and n-gram patterns suggest a strong emphasis on day-to-day employment issues and practical user concerns, showing that this dataset captures real usage patterns and frequently asked questions by workers and employers.

Jointplot (Scatter or KDE) showing relationship between Question vs Answer length



Comparison:

Comparison Overview:

We compared our four datasets (Qiwa FAQ, Qiwa Articles, BOE Official Law, and Istitlaa Public Consultation) based on text length, topic frequency, and content structure.

Interpretation of table Comparison:

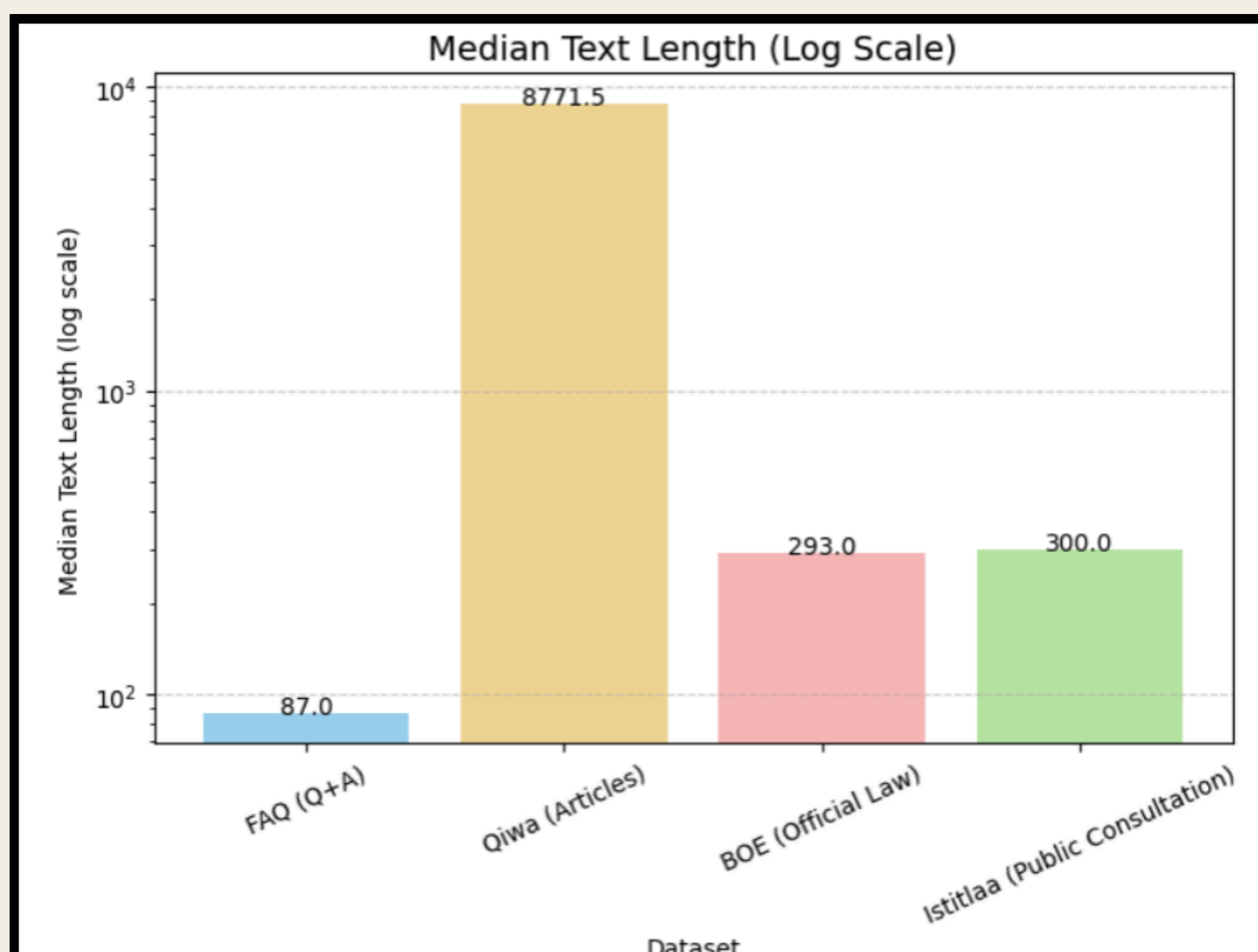
The comparison across the four datasets reveals major differences in both text length and topic focus. Qiwa (FAQ) has 32 entries with short texts (average 122 characters), which is consistent with its Q&A structure providing direct, concise information. Qiwa (Articles) shows extremely long texts (average 12,262 characters, median 8,771), reflecting detailed explanatory content written in full-article form. BOE (Official Law) includes 248 entries with moderate text length (average 426 characters), as it represents segmented legislative articles. Istitlaa (Public Consultation) contains 16 entries with medium-length proposals (average 522 characters), showing a balance between brevity and detail for public consultation. In terms of topics, all datasets share the recurring word (work), emphasizing labor-related content. However, Qiwa and BOE show a strong dominance of legal and procedural terms such as (article), while FAQ focuses on practical instructions and employee rights. Istitlaa highlights consultation-oriented terms like (training) and (modification), aligning with its goal of collecting public input on legal amendments. Overall, the results show a natural hierarchy from concise user-focused content (FAQ) to highly detailed official texts (Qiwa), reflecting the nature and purpose of each platform.

Comparison:

Interpretation of chart Comparison:

The chart shows a large difference in median text length across datasets. Qiwa has a significantly higher median due to its full-length articles, while FAQ, BOE, and Istitlaa contain much shorter texts. This confirms that Qiwa provides detailed explanatory content, whereas the other datasets focus on more concise, structured information.

	Dataset	Rows	Avg_Text_Length	Median_Text_Length	Top_Words
0	FAQ (Q+A)	32	122.88	87.0	العمل (7); يجب (4); حَقْد (4); علي (4); يحق (4)
1	Qiwa (Articles)	16	12262.50	8771.5	العمل... (946); المادة (738); علي (587); بعد (470)
2	BOE (Official Law)	248	426.79	293.0	العمل... (387); علي (339); بعد (292); المائتين (2)
3	Istitlaa (Public Consultation)	16	522.44	300.0	التدريب (41); المادة (36); العمل (28); علي (21)



Comparison:

Lexical Focus Comparison Analysis

The analysis was conducted to understand the function of each source's content, rather than simply looking at raw word frequency. The main idea was to define three lexical categories and calculate their concentration percentage (relative to total clean words) for each data source.

1. Lexical Categories

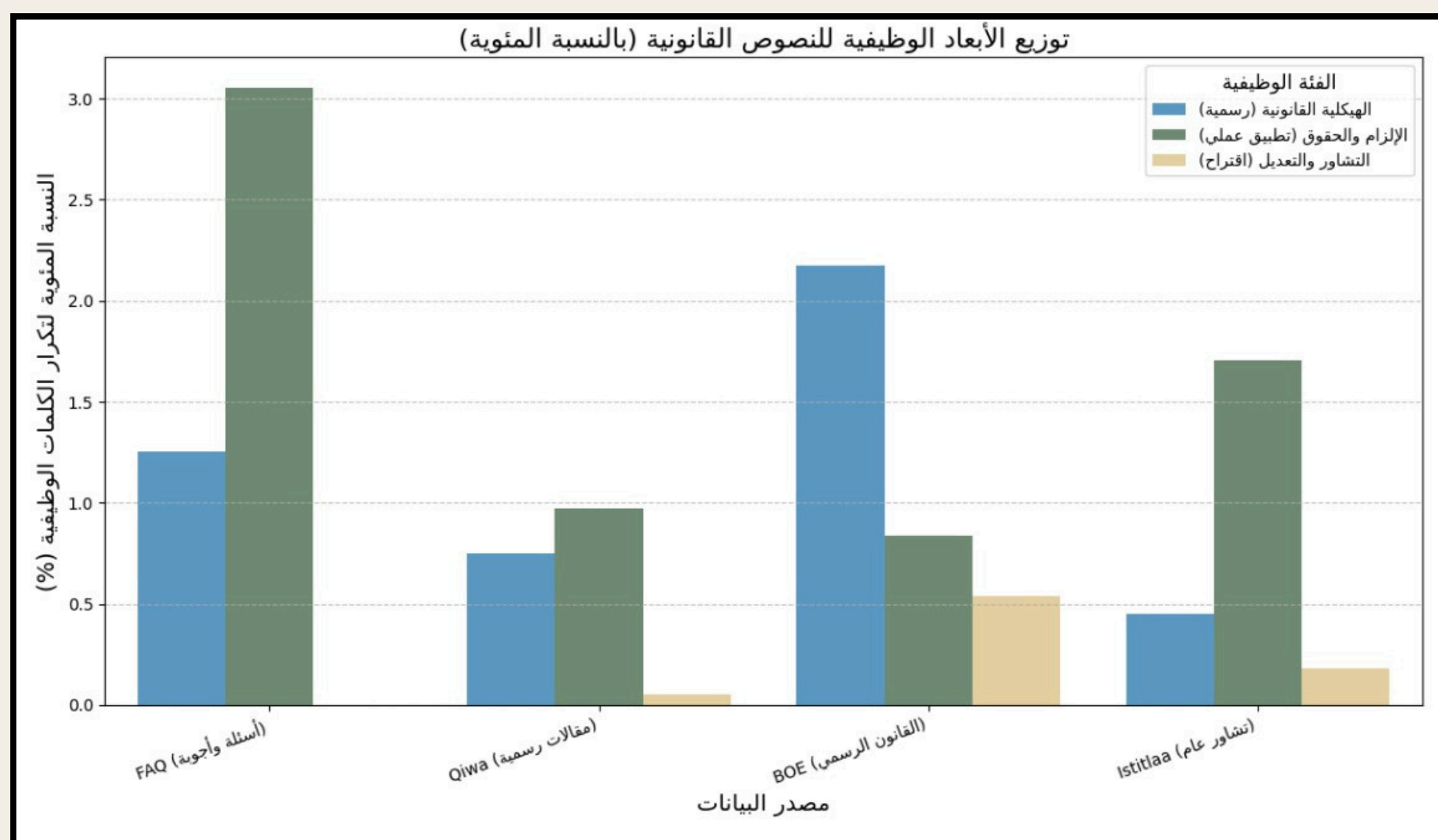
Legal terms were divided into three main categories based on the purpose of the text:

Legal Structure: Words indicating formality and organization, e.g., Article, Paragraph, Chapter, Section, Law, Decree.

Obligation & Rights: Words indicating practical legal application, e.g., must, obliged, entitled, for the worker, resignation, leave.

Consultation & Amendment: Words related to proposed changes or public opinion, e.g., proposed, amendment, survey, opinion, repeal, addition.

	Dataset	Legal_Structure	Obligation_Rights	Consultation_Amendment
0	FAQ (Q+A)	1.257	3.052	0.000
1	Qiwa (Articles)	0.747	0.973	0.050
2	BOE (Official Law)	2.172	0.840	0.536
3	Istitlea (Consultation)	0.449	1.706	0.180



Comparison:

a. Official Law (BOE):

Highest focus on Legal Structure (2.172%).

Represents raw and organized legal text. This source is the primary reference for citing legal articles in answers.

b. FAQs:

Highest focus on Obligation & Rights (3.052%).

Designed to answer user questions like “What must I do?” and “What are my rights?”

This source is key for training the model in practical, applied language.

c. Public Consultation (Istitlaa):

Highest focus on Consultation & Amendment (0.180%), though overall percentages are low.

Provides the model with historical context, showing that laws may undergo review and amendments, helping the system understand that some provisions are proposed changes.

d. Qiwa Articles:

Lowest concentration across the three categories.

Serves as explanatory or simplified material for legal texts. Low formal/legal focus makes it useful for training the model in simplified legal language.

3. Strategic Importance

This analysis emphasizes the need for multiple sources:

BOE ensures answers are grounded in the official reference (structure).

FAQ ensures answers are phrased in practical, user-friendly language.

This approach enabled the creation of a comprehensive, multidimensional Corpus of Law, supporting the training of an accurate and reliable legal system.

Summary of new insights and hypotheses:

New Insights:

- The datasets vary significantly in text length and content structure, reflecting their distinct purposes.
- Qiwa (FAQ) provides concise, user-oriented content designed for quick information retrieval.
- Qiwa (Articles) contains much longer and more formal texts, indicating its role as an official platform for detailed regulations.
- BOE (Official Law) includes moderately long legal articles, presenting structured and segmented legislative content.
- Istitlaa (Public Consultation) shows medium-length texts focused on proposed changes and public feedback.

Common topics across all datasets revolve around labor laws, regulations, and rights, emphasizing the shared domain of employment and governance. However, the frequency and depth of certain terms (like “Article” and “work”) highlight how each platform targets a different audience — from general users to policymakers.

Hypotheses:

- The difference in text length may correlate with the target audience, shorter texts for the public (FAQ) and longer ones for legal experts (Qiwa, BOE).
- The dominance of certain keywords (e.g., “work”) suggests a strong thematic overlap, indicating potential for cross-dataset topic alignment in future analysis.
- Public consultation texts (Istitlaa) may serve as a bridge between official laws (BOE) and user interpretations (FAQ/Qiwa).

Phase3:

Modelling and Evaluation

Modelling Approach

The goal of this phase was to evaluate different prompting strategies for generating legally accurate answers to questions related to the Saudi Labor Law. Since the task is not a traditional supervised classification or regression problem, we adopted a modelling strategy based on retrieval and generative large language models (LLMs).

We evaluated four main configurations:

1. Baseline Model – BM25 Retrieval

A keyword-based retrieval model using the BM25 algorithm was implemented as the baseline.

This model does not generate new answers; instead, it retrieves the most relevant article from the legal corpus.

A unified corpus was built from:

- BOE cleaned articles (official law)
- Istitlaa legal drafts
- Qiwa structured labor data
- Labor Law FAQ dataset

All texts were tokenized and indexed, and the baseline returned the top 3 retrieved articles for every test question.

This serves as a lower-bound benchmark for comparison with LLM-based models.

2. Zero-Shot Prompting

In this configuration, the LLM (Gemini 2.5 Pro) was asked to answer legal questions without seeing any examples.

This tests the model's general legal reasoning abilities.

The model generated 267 Q/A pairs, which were later evaluated for similarity, correctness, and linguistic quality.

3. One-Shot Prompting

Here, the LLM received one example Q/A pair before answering.

The example provided a structural and stylistic reference, helping the model understand:

- How legal answers should be formatted
- How to cite articles
- How to respond in a clear and formal tone

This resulted in more structured and grammatically polished answers.

A total of 207 Q/A pairs were generated.

4. Three-Shot Prompting

This configuration used three example Q/A pairs (few-shot prompting).

Providing multiple examples significantly improved:

- Legal grounding
- Consistency
- Accuracy when citing articles
- Handling of numerical rules (deductions, deadlines, penalties)

A total of 426 Q/A pairs were produced.

This model delivered the most stable and legally reliable answers.

Evaluation Methodology

A mixed evaluation approach was used, combining objective metrics, manual scoring, and qualitative analysis.

Objective (Automated) Metrics

We computed several automated metrics by comparing the generated answers with the official BOE article text:

1. Jaccard Overlap

Measures word-level similarity between the answer and the article.

2. Coverage (Answer in Article)

Percentage of answer tokens that also appear in the article text.

Higher values represent stronger grounding in the legal text.

3. Coverage (Article in Answer)

Percentage of article tokens represented inside the generated answer.

4. Relevance Score

Percentage of answers achieving minimum similarity thresholds (e.g., Jaccard ≥ 0.12 or Coverage ≥ 0.30).

5. Answer Length

Used to check whether answers were:

- Too short $\rightarrow$ missing legal details
- Too long $\rightarrow$ hallucinating or adding commentary

Manual (Human) Evaluation

For each generative model (Zero-Shot, One-Shot, Three-Shot), 50 Q/A pairs were manually reviewed and scored using:

Correctness (0–2)

- 0 = incorrect
- 1 = partially correct
- 2 = fully correct & aligned with the law

Grammar (1–5)

Evaluates quality of Arabic writing.

Style (0–2)

Measures clarity, formality, and consistency with legal writing.

Special attention was given to articles with numerical or procedural rules, such as:

- Articles 78–83 (disciplinary penalties)
- Articles 98–100 (wage deductions)
- Article 5 (applicability of the law)
- Article 97 (method of wage payment)

Evaluation Results

Objective Metrics Summary

Model	Avg Jaccard	Avg Coverage (Answer in Article)	Performance Summary
Baseline (BM25)	0.05	0.06 (best top-3 ≤ 0.077)	Very low similarity; retrieves raw articles only
Zero-Shot	~0.14	~0.32	Moderate similarity; unstable
One-Shot	~0.14	~0.32	Good structure; better style clarity
Three-Shot	0.13	0.39	Best grounding; most legally aligned

Key Insight:

Three-Shot had the highest Coverage, meaning its answers were most consistent with the content of the real article.

Evaluation Results

Manual Evaluation Summary

Model	Correctness	Grammar	Style	Notes
Zero-Shot	Lowest	Medium	weak	Often vague; lacks article citations
One-Shot	Good	Best	strong	Clear writing; sometimes verbose
Three-Shot	Highest	very good	Most consistent	Strongest legal accuracy; best handling of numeric rules

Key Insight:

Three-Shot consistently scored the highest in Correctness, especially for articles involving:

- Deadlines
- Penalties
- Numeric thresholds
- Wage calculations

Final Model Selection

Based on all evaluations:

Selected Model: Three-Shot Prompting

It provides the best balance between:

- Accuracy
- Stability
- Legal precision
- Coverage
- Style consistency

It is the most reliable model for building a Saudi Labor Law Q/A assistant.